

Project Trishul: AI-Powered Real-Time Landslide Early Warning & Disaster Management System (NER)

1. System Architecture Overview

```
graph TD
    subgraph ClientLayer ["Frontend Client (React + Tailwind + Vite + PWA)"]
        Landing["Public Landing Page & Citizen Portal"]
        UserPortal["Citizen / Field App (Offline sync, Geo-reports, SOS)"]
        AdminDashboard["Authority Command Center (GIS, Heatmaps, Analytics, Alerts)"]
    end

    subgraph APILayer ["Backend Services (Node.js + Express)"]
        AuthService["Auth & RBAC (JWT, Citizen, Field Officer, Admin)"]
        ReportService["Crowdsourced Incident & Media Service"]
        GISService["GeoJSON & Spatial Query Service"]
        AlertService["Multilingual Alert & Broadcast Engine (SMS / WebPush / WhatsApp)"]
        AIProxyService["AI Inference Gateway & IMD Ingestion"]
    end

    subgraph DataLayer ["Databases & Storage"]
        MongoDB["MongoDB with 2dsphere GeoJSON Indexes"]
        Cloudinary["Object Storage (Geo-tagged Evidence Photos/Videos)"]
        IndexedDB["Browser Offline Storage / IndexedDB"]
    end

    subgraph AIEngine ["AI/ML Analytics (FastAPI / Microservice)"]
        LandslideRiskModel["Slope Stability & Rain Threshold Model (XGBoost/LSTM)"]
        VisionHazardModel["Crack & Landslide Detection (YOLO / Vision CNN)"]
        WeatherFeed["IMD Weather & Satellite Data Feeder"]
    end

    ClientLayer <--> APILayer
    UserPortal <--> IndexedDB
    APILayer <--> DataLayer
    APILayer <--> AIEngine
```

2. Technology Stack

Frontend (Phase 1 - Current Focus)

- **Framework & Build Tool:** React 18 / 19 with Vite (Lightning fast HMR & minimal bundle).
- **Styling:** Tailwind CSS + Lucide-React icons + Framer Motion (for smooth micro-interactions & alert animations).
- **GIS & Interactive Mapping:**
 - **Leaflet / React-Leaflet** with OpenStreetMap, Mapbox, and Bhuvan/ISRO satellite tile layers.
 - Heatmap layers (`leaflet.heat`) for real-time landslide risk visualization.
 - GeoJSON rendering for roads, vulnerable bridges, and isolated village boundaries.

- **Charts & Data Visualization:** Recharts / Chart.js for rainfall trends, soil moisture gauge charts, and risk severity indexes.
- **Offline & PWA Support:**
 - Progressive Web App (Service Worker + Cache API).
 - `Dexie.js` / IndexedDB for offline geotagged incident draft storage with auto-sync when online.
- **State Management & Routing:** TanStack Query / React Context + React Router v6.
- **Multilingual Support:** `i18next` supporting English, Hindi, Assamese, Bengali, Mizo, Manipuri, etc.

Backend & Database (Phase 2 - Upcoming)

- **Runtime & Server:** Node.js with Express.js (Modular MVC architecture).
- **Database:** MongoDB Atlas with native **2dsphere** geospatial indexing for lightning-fast radius searches (`$nearSphere` , `$geoWithin`).
- **Real-Time Communication:** Socket.io (for live incident broadcasting to admin dashboards and emergency sirens).
- **Messaging / SMS:** Twilio / Fast2SMS / Firebase Cloud Messaging (Web Push & emergency notifications).
- **Media Storage:** Cloudinary / AWS S3 for geo-tagged image uploads with EXIF metadata parsing.

AI / Data Feeds (Future Integration)

- Python (FastAPI / Flask) running rainfall threshold prediction models (Antecedent Rainfall Index) and crack detection models.
- OpenWeather / IMD Weather API integrations.

3. Page-by-Page Feature Specifications

A. Front / Public Landing Page (/)

Purpose: Inform the public, provide quick emergency action buttons, live regional status summary, and raise awareness.

- Hero Section & Live Threat Ticker:**
 - Dynamic banner with current threat level across 8 North-Eastern states (Assam, Meghalaya, Sikkim, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Tripura).
 - High-impact CTA: "**Report Hazard / Blocked Road**" and "**Emergency SOS / Safe Route Finder**".
- Interactive Quick-Look Threat Map:**
 - Light interactive map showing active red/yellow alert zones across the NER.
- Live Weather & Disaster Advisories:**
 - IMD rainfall warning badges (Red/Orange/Yellow alert status).
- Key Pillars of Trishul:**
 - Explanation of AI Predictive early warning, Crowdsourced Field Reporting, Community offline sync, and Road Connectivity Monitoring.
- Emergency Helpline Directory:**
 - One-touch call buttons for SDRF, NDRF, District Disaster Management Authorities (DDMA), and Police.
- Multilingual Switcher & Voice Reader:**
 - Quick toggle between regional languages for accessibility.

B. Citizen & Field Officer Portal (/report & /citizen)

Purpose: Empower local citizens and field officials to report ground realities, check safe routes, and receive localized alerts even in low/no network.

1. One-Tap Geotagged Incident Reporter:

- Photo/video upload with automatic GPS capture, timestamp, and terrain tagging.
- Hazard categorization: *Road Blockage, Slope Movement, Landslide, Active Mudflow, Structural Cracks, Flash Flood.*
- Offline Queue: If connectivity drops, reports save locally in IndexedDB and automatically sync when connectivity resumes.

2. Road Connectivity & Evacuation Safe-Route Navigator:

- Search source & destination to check if mountain highways (e.g., NH-29, NH-10) are clear or blocked.
- Alternative safe-route recommendations avoiding high-slope hazard zones.

3. Local SOS & Safe Check-In:

- "I Am Safe" broadcast to family/local authorities.
- Offline SMS fallback for emergency distress signal with coordinates.

4. Community Alerts Feed:

- Localized timeline of verified incidents and government notices.

 C. Administrator & Disaster Management Command Center (/admin)

Purpose: Central command hub for District Magistrates, SDRF/NDRF, and BRO (Border Roads Organisation).

1. Comprehensive GIS Tactical Map:

- Multi-layer controls:
 -  Landslide Risk Heatmap (AI-calculated using slope, rainfall, soil moisture).
 -  Live Road Blockages & Infrastructure Vulnerability.
 -  Citizen-reported incidents (with unverified vs. verified statuses).
 -  Critical assets (Helipads, Relief camps, Hospitals, Equipment depots).

2. Incident Triage & Verification Workflow:

- Review incoming crowdsourced reports with AI duplicate detection and image crack analysis score.
- Actions: Verify, Dispatch Quick Response Team (SDRF/BRO), Issue Evacuation Order, Mark Resolved.

3. AI Risk Prediction & Early Warning Studio:

- 72-Hour Rain vs. Landslide probability forecast curve.
- Sensor telemetry dashboard (Piezo-sensors, Soil moisture, Rain gauges).

4. Emergency Broadcast & Siren Control:

- Geo-fenced broadcast creator: Select a polygon on the map -> Compose alert -> Send Multilingual SMS/Push alerts to all users inside that polygon.

5. Logistics & Resource Allocation Tracker:

- Track available excavators, JCBs, ambulances, and relief supplies near isolated villages.

4. Innovative Value-Adds for SIH (Smart India Hackathon)

1. Offline-First PWA (Service Workers + IndexedDB):

- Essential for NER hills where landslides usually knock down cellular towers.

2. AI Crowdsourced Deduplication & Confidence Scoring:

- Automatically clusters reports occurring within 100m of each other to avoid spamming the admin.

3. Isolated Village Resilience Score:

- Computes which remote village is at risk of total cutoff if a single mountain pass collapses.

4. Dual Map Engine (2D GIS + Satellite Switcher):

- Provides clear topological insights into steep slope gradients.
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5. Phased Implementation Plan

- **Phase 1 (Now - Frontend UI & Interactive Prototyping):**

- Setup React + Tailwind + Vite project in `C:\Users\Lenovo\Desktop\Trishul`.
- Install icons (`lucide-react`), mapping (`leaflet` , `react-leaflet`), charts (`recharts`), routing, and animation libraries.
- Implement dynamic, realistic mock data for NER states, sensors, road statuses, and incident feeds.
- Build **Landing Page**, **Citizen Reporting & Safe Route Finder**, and **Admin Command Center**.
- Provide full offline report caching simulation and responsive design.

- **Phase 2 (After user review & break - Backend & Real Integration):**

- Initialize Node.js, Express, MongoDB schemas with GeoJSON indexes.
- Connect Auth, REST API endpoints, and Socket.io for live updates.